

# TE Curve Verification Pipeline Official Model Verification Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

## Executive Verdict

This update accepts the completed `wave 4.1` robust-loss dispersion-aware campaign into the official `TE Curve Verification Pipeline` offline verification matrix.

### Decision

- `wave 4 series` is verified as an exploratory robust-loss baseline.
- No `wave 4 series` candidate is promoted over the current accepted `TE Curve Verification Pipeline` leaders.
- The project continues to maintain three parallel best surfaces: `Fw` , `Bw` , and `global` .
- Within `wave 4 series` , the strongest forward candidate is `track2h_mae_robust_Fw` .
- Within `wave 4 series` , the strongest backward candidate is `track2h_smooth_l1_robust_Bw` .
- Within `wave 4 series` , the strongest global candidate is `track2h_mae_robust_global` .

## Source Package

This official report consolidates these refreshed artifacts:

- metric matrix:  
doc/reports/analysis/track2/Track 2 Directional Model Comparison.md ;
- matrix summary:  
output/validation\_checks/track2\_reference\_comparison/2026-06-11-15-04-04\_\_track2\_full\_directional\_family\_matrix\_track2h\_robust\_loss\_track2\_refresh\_2026\_06\_11/validation\_summary.yaml ;
- per-condition metrics:  
output/validation\_checks/track2\_reference\_comparison/2026-06-11-15-04-04\_\_track2\_full\_directional\_family\_matrix\_track2h\_robust\_loss\_track2\_refresh\_2026\_06\_11/per\_condition\_metrics.csv ;
- validation report:  
doc/reports/analysis/validation\_checks/track2/2026-06-11-15-14-38\_track2\_full\_directional\_family\_matrix\_track2h\_robust\_loss\_track2\_refresh\_2026\_06\_11\_report.md ;
- best-model collage report:  
doc/reports/analysis/track2/best\_model\_collage\_report/[2026-06-11]/track2\_best\_model\_collage\_report.md ;
- multi-model curve comparison report:  
doc/reports/analysis/track2/multi\_model\_curve\_comparison\_report/[2026-06-11]/track2\_multi\_model\_curve\_comparison\_report.md ;
- operator launch logs:  
output/validation\_checks/track2\_operator\_launch\_logs/2026-06-11-15-04-00\_track2h\_robust\_loss\_track2\_refresh\_2026\_06\_11/ .

## Candidate Refresh

The refresh added 9 registry-backed Wave 4 series candidates covering robust MAE , SmoothL1 , and LogCosh losses across the global , Fw , and Bw surfaces.

| Surface | Robust Loss | Candidate                       |
|---------|-------------|---------------------------------|
| global  | MAE         | track2h_mae_robust_global       |
| Fw      | MAE         | track2h_mae_robust_Fw           |
| Bw      | MAE         | track2h_mae_robust_Bw           |
| global  | SmoothL1    | track2h_smooth_l1_robust_global |
| Fw      | SmoothL1    | track2h_smooth_l1_robust_Fw     |
| Bw      | SmoothL1    | track2h_smooth_l1_robust_Bw     |
| global  | LogCosh     | track2h_log_cosh_robust_global  |
| Fw      | LogCosh     | track2h_log_cosh_robust_Fw      |
| Bw      | LogCosh     | track2h_log_cosh_robust_Bw      |

The matrix source group is track2h\_dispersion\_aware\_modeling\_registry . The matrix now contains 141 candidates.

## Verification Rules

| Surface | Training or Archive Scope              | Evaluation Scope                                    |
|---------|----------------------------------------|-----------------------------------------------------|
| Fw      | forward-only training or archive       | forward TE Curve Verification Pipeline curves only  |
| Bw      | backward-only training or archive      | backward TE Curve Verification Pipeline curves only |
| global  | forward and backward training together | both directions, reported by direction and combined |

The `global` , `Fw` , and `Bw` branches are carried forward in parallel. They are not collapsed into one scalar winner.

## Current Leaders

### Overall Direction Leaders

| Direction       | Current Strongest Candidate               | MAE [deg] | RMSE [deg] | Mean [%] | P95 [%] |
|-----------------|-------------------------------------------|-----------|------------|----------|---------|
| forward         | <code>rcim_retuned_GBM19_Fw</code>        | 0.001089  | 0.001299   | 2.372    | 4.912   |
| backward        | <code>periodic_gru_sequence_Bw</code>     | 0.002392  | 0.002639   | 5.466    | 14.820  |
| global combined | <code>periodic_gru_sequence_global</code> | 0.002704  | 0.002949   | 6.139    | 13.200  |

### Repository-Owned Static Baselines

| Direction       | Current Strongest Candidate | MAE [deg] | RMSE [deg] | Mean [%] | P95 [%] |
|-----------------|-----------------------------|-----------|------------|----------|---------|
| backward        | <code>tree_Bw</code>        | 0.003258  | 0.003651   | 7.051    | 14.116  |
| global combined | <code>tree_global</code>    | 0.003144  | 0.003533   | 6.854    | 13.314  |

### Repository-Owned Neural Leaders

| Surface         | Current Strongest Candidate               | MAE [deg] | RMSE [deg] | Mean [%] | P95 [%] |
|-----------------|-------------------------------------------|-----------|------------|----------|---------|
| Bw              | <code>periodic_gru_sequence_Bw</code>     | 0.002392  | 0.002639   | 5.466    | 14.820  |
| global combined | <code>periodic_gru_sequence_global</code> | 0.002704  | 0.002949   | 6.139    | 13.200  |

## Wave 4 series Result

| Surface | Candidate                                | Robust Loss           | MAE [deg] | RMSE [deg] | Mean [%] | P95 [%] |
|---------|------------------------------------------|-----------------------|-----------|------------|----------|---------|
| Bw      | <code>track2h_smooth_l1_robust_Bw</code> | <code>SmoothL1</code> | 0.003078  | 0.003403   | 6.676    | 15.410  |
| Fw      | <code>track2h_mae_robust_Fw</code>       | <code>MAE</code>      | 0.003134  | 0.003382   | 6.956    | 12.470  |
| Fw      | <code>track2h_smooth_l1_robust_Fw</code> | <code>SmoothL1</code> | 0.003300  | 0.003545   | 7.342    | 13.401  |

| Surface         | Candidate                       | Robust Loss | MAE [deg] | RMSE [deg] | Mean [%] | P95 [%] |
|-----------------|---------------------------------|-------------|-----------|------------|----------|---------|
| Fw              | track2h_log_cosh_robust_Fw      | LogCosh     | 0.003344  | 0.003595   | 7.427    | 12.598  |
| global combined | track2h_mae_robust_global       | MAE         | 0.003401  | 0.003715   | 7.504    | 13.873  |
| global combined | track2h_smooth_l1_robust_global | SmoothL1    | 0.003417  | 0.003719   | 7.539    | 14.332  |
| Bw              | track2h_mae_robust_Bw           | MAE         | 0.003433  | 0.003750   | 7.506    | 14.575  |
| Bw              | track2h_log_cosh_robust_Bw      | LogCosh     | 0.003486  | 0.003811   | 7.628    | 12.872  |
| global combined | track2h_log_cosh_robust_global  | LogCosh     | 0.003498  | 0.003819   | 7.697    | 14.405  |

Robust losses improve the previous Wave 3.3 branch on the best Fw and Bw surfaces. The strongest Fw Wave 4 series candidate improves over the best Wave 3.3 Fw candidate from 0.003174 to 0.003134 MAE. The strongest Bw Wave 4 series candidate improves over the best Wave 3.3 Bw candidate from 0.003436 to 0.003078 MAE. The best Wave 4 series global candidate remains slightly behind the best Wave 3.3 global candidate.

### Visual Evidence

The 2026-06-11 collage and overlay bundles were regenerated with dedicated Wave 4 series sections:

- Forward Wave 4.1 Robust-Loss Models ;
- Backward Wave 4.1 Robust-Loss Models ;
- Global Wave 4.1 Robust-Loss Models in the collage report;
- Forward Reference Tree And Wave 4 series Overlay ;
- Backward Reference Tree And Wave 4 series Overlay .

The visual package supports the matrix decision: robust losses are useful controls for dispersion-aware training, especially in the backward direction, but they do not yet replace the strongest periodic temporal branch.

### Campaign Update Ledger

| Date       | Campaign or Update                            | Candidate Scope                                                                                  | Matrix Status                          | Visual Status                                                             | Decision                                    |
|------------|-----------------------------------------------|--------------------------------------------------------------------------------------------------|----------------------------------------|---------------------------------------------------------------------------|---------------------------------------------|
| 2026-06-11 | Wave 4.1 robust-loss dispersion-aware refresh | 9 robust global , Fw , and Bw candidates                                                         | included in the 141 - candidate matrix | dated collage and overlay bundles regenerated with Wave 4 series sections | verified exploratory baseline; not promoted |
| 2026-06-10 | Wave 3.3 curve-aware training refresh         | 12 pointwise-control, centered-shape, offset, and full-composite global , Fw , and Bw candidates | included in the 132 - candidate matrix | dated collage and overlay bundles regenerated with Wave 3.3 sections      | verified exploratory baseline; not promoted |
| 2026-06-08 | Wave 3.2 harmonic-offset probe refresh        | 6 clean and harmonic global , Fw , and Bw candidates plus 3 rechecked Wave 3.1 candidates        | included in the 120 - candidate matrix | dated collage and overlay bundles regenerated with Wave 3.2 sections      | verified exploratory baseline; not promoted |

| Date       | Campaign or Update                                 | Candidate Scope                                                              | Matrix Status                          | Visual Status                                                | Decision                                                                                                       |
|------------|----------------------------------------------------|------------------------------------------------------------------------------|----------------------------------------|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| 2026-06-04 | Wave 3.1 offset-aware probe refresh                | 3 global , Fw , and Bw sequential residual offset candidates                 | included in the 114 - candidate matrix | dated collage and overlay bundles regenerated                | verified exploratory baseline; not promoted                                                                    |
| 2026-05-28 | Wave 2.3 residual harmonic temporal hybrid refresh | 18 global , Fw , and Bw residual harmonic GRU/LSTM candidates                | included in the 111 - candidate matrix | collage and overlay reports refreshed with Wave 2.3 sections | verified exploratory baseline; not promoted over Wave 2.2 or accepted TE Curve Verification Pipeline baselines |
| 2026-05-26 | Wave 2.2 harmonic temporal hybrid refresh          | periodic temporal convolution, GRU, and LSTM global , Fw , and Bw candidates | included                               | collage and overlay reports refreshed                        | strongest repository-owned neural branch                                                                       |

## Closeout Decision

Wave 4 series does not change the accepted TE Curve Verification Pipeline baseline. The current direction-parallel decision remains:

- Fw : rcim\_retuned\_GBM19\_Fw remains the strongest overall forward candidate; track2h\_mae\_robust\_Fw is the strongest Wave 4 series forward candidate and is slightly better than the best Wave 3.3 forward candidate.
- Bw : periodic\_gru\_sequence\_Bw remains the strongest practical repository-owned backward candidate; track2h\_smooth\_11\_robust\_Bw is the strongest Wave 4 series candidate and improves over the static tree\_Bw baseline in curve-verification MAE.
- global : periodic\_gru\_sequence\_global remains the strongest repository-owned bidirectional neural candidate; within Wave 4 series, the MAE robust global candidate is the strongest but remains behind the current Wave 3.3 global leader and the accepted periodic temporal global branch. The next modeling step should keep robust losses as a validated ingredient, but not treat them as sufficient. The evidence supports preparing the next dispersion-aware package around quantile or probabilistic regression, followed by mixture-density heads and latent-state or hysteresis-aware variants before the integrated multi-task / multi-head model branch.